

Advance Microeconomic Theory

Homework - 02

Date: 14/11/2024

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Session: 2018-19

Ans: to the Q: no: 01

To check if a utility function is homogenous, we test whether scaling all inputs by the same factor, say k (where $k > 0$), results in the function scaling by a constant power.

Now, we check the homogeneity of the given function.

(a) $u(x, y) = (xy)^4$

$$\begin{aligned} \therefore u(kx, ky) &= \{(kx)(ky)\}^4 \\ &= (k^2xy)^4 \\ &= k^4(xy)^4 \end{aligned}$$

$\therefore$ The function is homogenous of degree 4.

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(b) $U(x, y) = (xy)^{1/2}$

$$\begin{aligned}\therefore U(kx, ky) &= \{(kx)(ky)\}^{1/2} \\ &= (k^2 xy)^{1/2} \\ &= k(xy)^{1/2}\end{aligned}$$

$\therefore$ The function is homogeneous with degree 1.

(c) $U(x, y) = \min(x, 2y)$

$$\therefore U(kx, ky) = \min(kx, 2ky)$$

In the above function, x and y are complementary goods and they are consumed together in fixed proportion ($x=2y$), and an increase in the availability of one good without a corresponding increase in other does not increase utility.

Hence we can factor out k , and write,

$$U(kx, ky) = k \min(x, 2y)$$

$\therefore$ The function is homogeneous of degree 1.

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(d) $U(x, y) = 2x + y$

$$\therefore U(kx, ky) = 2kx + ky \\ = k(2x + y)$$

$\therefore$ The function is homogeneous of degree 1.

(e) $U(x, y) = x^r y^r + xy$

$$\therefore U(kx, ky) = (kx)^r (ky)^r + (kx)(ky) \\ = k^4 x^r y^r + k^2 xy$$

As the terms scale differently (k^4 and k^2)

the function is not homogeneous.

Ans: to the Q: no: 02

In our previous analysis, all functions except function (e) proved to be homogenous.

As all homogenous function are homothetic function, we only need to check whether the function (e) is a homothetic function.

$$(e) \quad u(x, y) = x^2 y^2 + xy$$

For a homothetic function, the marginal rate of substitution (MRS) depends only on the ratio of the amount of the two goods, instead of the total quantities of goods.

$$\text{We know, } MRS = \frac{\partial u / \partial x}{\partial u / \partial y}$$

From function (e),

$$\frac{\partial u}{\partial x} = 2xy^2 + y$$

$$\frac{\partial u}{\partial y} = 2x^2y + x$$

$$\text{Now, } MRS = \frac{\partial u / \partial x}{\partial u / \partial y} = \frac{2xy^2 + y}{2x^2y + x}$$

$$= \frac{y(2xy + 1)}{x(2xy + 1)}$$

$$= y/x \Rightarrow \text{A.d.O}$$

Thus, the MRS of function (e) depends only on the ratio of good x and y . Therefore, it is a homothetic function despite not being homogenous.

From Equation (i) & (ii), we get,

$$\frac{\lambda P_x}{x P_y} = \frac{y^{1/2} / 2x^{1/2}}{x^{1/2} / 2y^{1/2}}$$

$$\Rightarrow \frac{P_x}{P_y} = \frac{y}{x}$$

$$\therefore y = \frac{P_x x}{P_y}$$

Ans: to the Q: no: 03

(a)

Given that $u = (xy)^r$

To find out the Marshallian demands and indirect utility function, we need to optimize the function subject to one's budget constraint.

$\therefore$ We can write the problem as,

$$\text{Max } u = (xy)^r$$

$$\text{subject to, } p_x x + p_y y = 1$$

Now, the Lagrangian function will be,

$$\text{Max}_{x, y, \lambda} \mathcal{L} = (xy)^r + \lambda [1 - p_x x - p_y y]$$

According to the first order condition,

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy^r - \lambda p_x = 0 \quad \text{--- (i)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2x^r y - \lambda p_y = 0 \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial x} = I - P_x x - P_y y = 0 \quad \text{--- (iii)}$$

From equation (i) and (ii), we get,

$$\frac{P_x}{P_y} = \frac{2x y r}{2x^2 y}$$

$$= \frac{y}{x}$$

$$\therefore y = \frac{P_x}{P_y} \cdot x \quad \text{Similarly} \quad x = \frac{P_y}{P_x} \cdot y \quad \text{--- (iv)}$$

Substituting, ~~$x = \frac{P_y}{P_x} \cdot y$~~ $\left[y = \frac{P_x}{P_y} \cdot x \right]$ in equation (iii), we get,

$$I = P_x x + P_y \cdot \frac{P_x}{P_y} \cdot x$$

$$\Rightarrow I = 2P_x x$$

$$\therefore x^* = \frac{I}{2P_x} \quad \text{Similarly,} \quad y^* = \frac{I}{2P_y}$$

$\therefore$ The Marshallian demand functions are,

$$x^* = \frac{I}{2P_x} \quad \text{and} \quad y^* = \frac{I}{2P_y}$$

Now, the indirect utility function,

$$\begin{aligned} V(P_1, P_2, I) &= \left(\frac{I}{2P_1} \cdot \frac{I}{2P_2} \right)^{1/2} \\ &= \left(\frac{I^2}{4P_1 P_2} \right)^{1/2} \\ &= \frac{I^2}{16 P_1^{1/2} P_2^{1/2}} \end{aligned}$$

$\therefore$ The Indirect utility function is $\frac{I^2}{16 P_1^{1/2} P_2^{1/2}}$

(b)
Here, Max $U = (xy)^{1/2}$

s.t. $P_1 x + P_2 y = I$

thus

$$\text{Max}_{x, y, \lambda} L = x^{1/2} y^{1/2} + \lambda [I - P_1 x - P_2 y]$$

According to the first order condition,

$$\frac{\partial L}{\partial x} = \frac{1}{2} x^{-1/2} y^{1/2} - \lambda P_1 = 0 \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2} - \lambda P_2 = 0 \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_1 x - P_2 y = 0 \quad \text{--- (iii)}$$

From Equation (i) & (ii), we get,

$$\frac{\lambda p_x}{\lambda p_y} = \frac{y^{1/2} / 2x^{1/2}}{x^{1/2} / 2y^{1/2}}$$

$$\Rightarrow \frac{p_x}{p_y} = \frac{y}{x}$$

$$\therefore y = \frac{p_x x}{p_y} \quad \text{--- (iv)}$$

Substituting the value of y in (iii)

$$I = p_x x + p_y \cdot \frac{p_x x}{p_y}$$

$$\Rightarrow 1 = 2p_x x$$

$$\therefore x^* = \frac{1}{2p_x}$$

Substituting the value of x^* in (iv),

$$y^* = \frac{p_x}{p_y} \cdot \frac{1}{2p_x} = \frac{1}{2p_y}$$

$\therefore$ The Marshallian demands are,

$$x^* = \frac{1}{2p_x} \quad \text{and} \quad y^* = \frac{1}{2p_y}$$

Now, the indirect utility function -

$$V(P_x, P_y, I) = \left(\frac{I}{2P_x}, \frac{I}{2P_y} \right)^{1/2}$$

$$= \left(\frac{I^2}{4P_x P_y} \right)^{1/2}$$

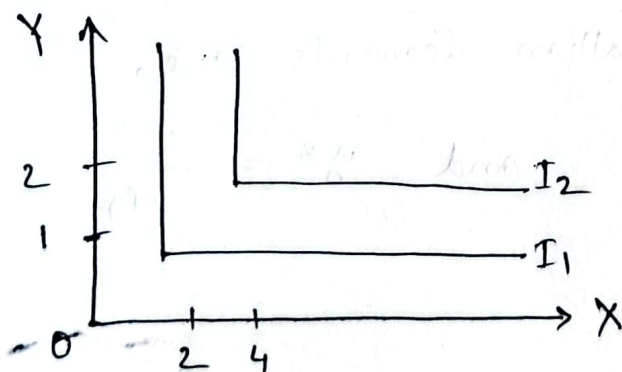
$$= \frac{I}{2\sqrt{P_x P_y}}$$

$\therefore$ The indirect utility function is $\frac{I}{2\sqrt{P_x P_y}}$

(C)

$$u = \min(x, 2y)$$

In this case, the indifference curve is L-shaped and x and y must be consumed in fixed proportion. The graphical representation will be;



In this situation, a utility maximizing consumer will choose such that, $x = 2y$ — (i)

$x > 2y$ or $x < 2y$ will not be feasible case because the goods are complementary & consuming more of one good holding the other good constant will not increase utility.

Let the budget constraints;

$$I = P_x x + P_y y$$

$$= P_x x + P_y \cdot \frac{1}{2} x \quad [\text{From (i)}]$$

$$= x \left(P_x + \frac{1}{2} P_y \right)$$

$$\therefore x^* = \frac{I}{P_x + \frac{1}{2} P_y} = \frac{2I}{2P_x + P_y}$$

$$\text{Similarly, } y^* = \frac{I}{2P_x + P_y}$$

$\therefore$ The Marshallian demands are;

$$x^* = \frac{2I}{2P_x + P_y} \quad \text{and} \quad y^* = \frac{I}{2P_x + P_y}$$

Now, Indirect Utility Function -

As previously mentioned, good x and y complementary goods,

$$\text{Therefore, } V(P, I) = \min \left(\frac{2I}{2P_x + P_y}, \frac{2I}{2P_x + P_y} \right) \\ = \frac{2I}{2P_x + P_y}$$

(d)

$$U = 2x + y$$

$$\text{s.t. } P_x x + P_y y = I$$

Now, the primal problem is -

$$\text{Max}_{x, y, \lambda} \quad \mathcal{L} = 2x + y + \lambda [I - P_x x - P_y y]$$

According to the first order condition,

$$\frac{\partial \mathcal{L}}{\partial x} = 2 - \lambda P_x = 0 \quad \text{--- (i)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 1 - \lambda P_y = 0 \quad \text{--- (ii)}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_x x - P_y y = 0 \quad \text{--- (iii)}$$

From equation (i) & (ii), we get,

$$\frac{P_x}{P_y} = 2, \text{ Which implies that, the}$$

given function provides a corner solution

Now, from equation (iii),

$$\text{if, } y^* = 0, \text{ then, } I = P_x x \therefore x^* = I/P_x$$

$$\text{if, } x^* = 0, \text{ then, } I = P_y y \therefore y^* = I/P_y$$

$\therefore$ The Marshallian demands are,

$$x^* = I/P_x \text{ or } y^* = I/P_y$$

Now, the indirect utility function -

$$v(P_x, P_y, I) = \max \left(\frac{I}{P_x}, \frac{I}{P_y} \right)$$

(e)

Here, $\max u = x^{\frac{1}{2}}y^{\frac{1}{2}} + xy$.

s.t. $P_x x + P_y y = I$

Now, the primal problem will be,

$$\max_{x, y, \lambda} L = x^{\frac{1}{2}}y^{\frac{1}{2}} + xy + \lambda [I - P_x x - P_y y]$$

According to the first order condition,

$$\frac{\partial L}{\partial x} = \frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}} + y - \lambda P_x = 0 \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = \frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}} + x - \lambda P_y = 0 \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_x x - P_y y = 0 \quad \text{--- (iii)}$$

From equation (i) & (ii), we get,

$$\frac{\frac{1}{2}x^{-\frac{1}{2}}y^{\frac{1}{2}} + y}{\frac{1}{2}x^{\frac{1}{2}}y^{-\frac{1}{2}} + x} = \frac{\lambda P_x}{\lambda P_y}$$

$$\Rightarrow \frac{y(2x+1)}{x(2y+1)} = \frac{P_x}{P_y}$$

$$\therefore y = \frac{P_x}{P_y} \cdot x \quad \text{--- (iv)}$$

Substituting the value of y in equation (iii),

$$I = P_x x + P_y \cdot \frac{P_x}{P_y} \cdot x$$

$$\Rightarrow I = 2P_x x$$

$$\therefore x^* = \frac{I}{2P_x}$$

Again, substituting the value of x^* in equation (iv)

$$y^* = \frac{P_x}{P_y} \cdot \frac{I}{2P_x} = \frac{I}{2P_y}$$

$\therefore$ Marshallian demands are;

$$x^* = \frac{I}{2P_x} \quad \text{and} \quad y^* = \frac{I}{2P_y}$$

Now, the indirect utility function,

$$\begin{aligned} V(P_x, P_y, I) &= \left(\frac{I}{2P_x}\right)^{\alpha} \left(\frac{I}{2P_y}\right)^{\beta} + \frac{I}{2P_x} \cdot \frac{I}{2P_y} \\ &= \frac{I^{\alpha}}{4P_x^{\alpha}} \cdot \frac{I^{\beta}}{4P_y^{\beta}} + \frac{I^{\alpha}}{4P_x P_y} \\ &= \frac{I^{\alpha+\beta}}{16P_x^{\alpha} P_y^{\beta}} + \frac{I^{\alpha}}{4P_x P_y} \\ &= \frac{I^{\alpha}}{4P_x P_y} \left(\frac{I^{\beta}}{4P_x^{\alpha} P_y^{\beta}} + 1 \right) \end{aligned}$$

[This is the indirect utility function]

Ans: to the Q1 no: 04

(a)

Here, $u = (xy)^n$

$$\text{So, } MU_x = 2xy^{n-1}$$

$$MU_y = 2x^n y$$

$$\text{and, } \frac{\partial MU_x}{\partial x} = \frac{\partial^2 u}{\partial x^2} = 2y^{n-1} > 0$$

$$\frac{\partial MU_y}{\partial y} = \frac{\partial^2 u}{\partial y^2} = 2x^n > 0$$

$\therefore$ This function does not exhibit diminishing marginal utility.

Downward Slopping Marshallian Demand

From (3)-a, we get,

$$x^* = \frac{I}{2P_x} \quad \& \quad y^* = \frac{I}{2P_y}$$

If we differentiate the demand function with respect to their price, we will get the slope.

Now,

$$\frac{\partial x^*}{\partial p_x} = - \frac{1}{2p_x} < 0$$

$$\frac{\partial y^*}{\partial p_y} = - \frac{1}{2p_y} < 0$$

∴ The demand curve is downward sloping

(b)

Here, $U = (xy)^{1/2}$

$$\text{So, } MU_x = \frac{\partial U}{\partial x} = \frac{1}{2} x^{-1/2} y^{1/2}$$

$$MU_y = \frac{\partial U}{\partial y} = \frac{1}{2} x^{1/2} y^{-1/2}$$

$$\begin{aligned} \text{Now, } \frac{\partial MU_x}{\partial x} &= \frac{\partial^2 U}{\partial x^2} = -\frac{1}{4} x^{-3/2} y^{1/2} \\ &= - \frac{x^{-3/2} y^{1/2}}{4} \\ &= - \frac{y^{1/2}}{4x^{3/2}} < 0 \end{aligned}$$

$$\begin{aligned} \text{Again, } \frac{\partial MU_y}{\partial y} &= \frac{\partial^2 U}{\partial y^2} = -\frac{1}{4} x^{1/2} y^{-3/2} \\ &= - \frac{x^{1/2}}{4y^{3/2}} < 0 \end{aligned}$$

∴ This function exhibits Marginal Utility which is diminishing.

From (3)-b, we get,

$$x^* = \frac{I}{2P_x} \quad \& \quad y^* = \frac{I}{2P_y}$$

Differentiating marshallian demand with ~~respect to~~
respective prices;

$$\frac{\partial x^*}{\partial P_x} = - \frac{I}{2P_x^2}$$

$$\frac{\partial y^*}{\partial P_y} = - \frac{I}{2P_y^2} < 0$$

$\therefore$ Thus, the marshallian demands are ~~downward~~
~~are~~ downward sloping.

(d)

Here, $U = 2x + y$

∴ Marginal Utility of X, $MU_x = 2$

" " of Y, $MU_y = 1$

Since MU_x & MU_y both are constant, the function does not exhibit diminishing marginal utility.

Now, from (3)-d, we get,

$$x^* = \frac{1}{p_x + 2p_y} \quad \& \quad y^* = \frac{2I}{p_x + p_y}$$

Differentiating with respect to their prices;

$$\begin{aligned} \frac{\partial x^*}{\partial p_x} &= \frac{\partial}{\partial p_x} \left(\frac{1}{p_x + 2p_y} \right) \\ &= \frac{(p_x + 2p_y) - 1 \cdot 1}{(p_x + 2p_y)^2} \end{aligned}$$

$$= - \frac{1}{(p_x + 2p_y)^2} < 0$$

$$\begin{aligned}
 \text{And, } \frac{\partial x^*}{\partial p_z} &= \frac{b}{\partial p_z} \left(\frac{2I}{p_x + 2p_z} \right) \\
 &= \frac{0 \cdot (p_x + 2p_z) - 2I \cdot 2}{(p_x + 2p_z)^2} \\
 &= - \frac{4I}{(p_x + 2p_z)^2} < 0
 \end{aligned}$$

$\therefore$ The Marshallian demand is downward sloping.

(e)

Here, $U = xz^r + y$

$$\text{So, } MU_x = 2xz^{r-1} + y \quad \text{--- (i)}$$

$$MU_y = 2xz^r + y \quad \text{--- (ii)}$$

$$\text{And, } \frac{\partial MU_x}{\partial MU_y} = 2z^r > 0$$

$$\frac{\partial MU_x}{\partial z} = 2x^r > 0,$$

Thus, both x & z has increasing marginal ~~po~~ utility.

Now, from (3) - e, we get,

$$x^* = \frac{I}{2p_x} \quad \& \quad y^* = \frac{I}{2p_y}$$

Differentiating with respect to the relative prices;

$$\frac{\partial x^*}{\partial p_x} = -\frac{I}{2p_x^2} < 0$$

$$\text{and, } \frac{\partial y^*}{\partial p_x} = -\frac{I}{2p_x^2} < 0$$

$\therefore$ These Marshallian demands are ~~downward~~ downward-sloping.

Therefore, as we have seen in our example,

function (a), (d) and (e) do not exhibit diminishing marginal utility, yet they have downward sloping demand curve.

So, it is not necessary for the marginal utility to be diminishing, in order to get a downward sloping demand curve.

Ans: to the Q: no: 05

(b)

Given, $u = (xy)^{1/2}$

s.t. $p_x x + p_y y = I$

The dual problem will be,

Min $p_x x + p_y y$

s.t. $u = (xy)^{1/2} = \bar{u}$

Now,

Min $L = p_x x + p_y y + \lambda [\bar{u} - x^{1/2} y^{1/2}]$
 x, y, λ

First order condition requires;

$$\frac{\partial L}{\partial x} = p_x - \lambda \left(\frac{1}{2}\right) x^{-1/2} y^{1/2} = 0 \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial y} = p_y - \lambda \left(\frac{1}{2}\right) x^{1/2} y^{-1/2} = 0 \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = \bar{u} - x^{1/2} y^{1/2} = 0 \quad \text{--- (iii)}$$

From (i) & (ii) -

$$\frac{P_x}{P_y} = \frac{x^{-1/2} y^{1/2}}{x^{1/2} y^{-1/2}} = \frac{y}{x}$$

$$\therefore \cancel{x = \frac{P_y}{P_x} \cdot y} \quad x = \frac{P_y}{P_x} \cdot y, \quad y = \frac{P_x}{P_y} \cdot x \quad \text{--- (iv)}$$

Now, From (iii),

$$\begin{aligned} \bar{u} &= x^{1/2} y^{1/2} \\ &= \left(\frac{P_y}{P_x} \cdot y \right)^{1/2} y^{1/2} \quad [\text{From (iv)}] \\ &= \sqrt{\frac{P_y}{P_x}} \cdot y \quad (1) \end{aligned}$$

$$\therefore y^* = \bar{u} \sqrt{\frac{P_x}{P_y}}$$

$$\therefore h_y(p, \bar{v}) = \bar{v} \sqrt{\frac{P_x}{P_y}} \quad [\text{ Hicksian demand for good } y]$$

Again,

$$\begin{aligned} \bar{u} &= x^{1/2} y^{1/2} \\ &= x^{1/2} \left(\frac{P_x}{P_y} \cdot x \right)^{1/2} \\ &= \sqrt{\frac{P_x}{P_y}} \cdot x \end{aligned}$$

$$\therefore x^* = \bar{u} \sqrt{\frac{P_y}{P_x}}$$

$$\therefore h_x(p, \bar{v}) = \bar{u} \sqrt{\frac{P_y}{P_x}}$$

Now, optimal value function,

$$L^* = P_1 h_1(P, \bar{u}) + P_2 h_2(P, \bar{u}) + \lambda [\bar{u} - \{h_1(P, \bar{u})\}^{1/2} - \{h_2(P, \bar{u})\}^{1/2}]$$

$$= P_1 \cdot \bar{u} \sqrt{\frac{P_2}{P_1}} + P_2 \cdot \bar{u} \sqrt{\frac{P_1}{P_2}}$$

$$\therefore e(P, \bar{u}) = \bar{u} \sqrt{P_1 P_2} + \bar{u} \sqrt{P_1 P_2}$$

$$= 2\bar{u} \sqrt{P_1 P_2} \quad \text{[Expenditure Function]}$$

$$h_x = \frac{\partial e}{\partial P_x} = ?$$

$$h_y = \frac{\partial e}{\partial P_y} = ?$$

(e)

Given that,

$$u = \min(x, 2y)$$

$$\text{s.t. } P_1 x + P_2 y = I \quad \text{--- (i)}$$

The dual problem will be;

$$\text{Min } P_1 x + P_2 y$$

$$\text{s.t. } u = \min(x, 2y) = \bar{u} \quad \text{--- (ii)}$$

But from (i), we have already derived

indirect utility function, which is,

$$V(P, I) = \frac{2I}{2P_x + P_y}$$

therefore, the expenditure function is;

$$e(P, \bar{u}) = \left(P_x + \frac{P_y}{2}\right) \bar{u} \quad \text{--- (iii)}$$

By taking the derivative of (iii). with respect to ~~P~~ P_x , compensated demand function can be constructed. Therefore;

$$\frac{\partial e(P, \bar{u})}{\partial P_x} = \frac{\partial}{\partial P_x} \left\{ \left(P_x + \frac{P_y}{2}\right) \bar{u} \right\}$$

$$\therefore h_x(P, \bar{u}) = \bar{u} \quad \left[\begin{array}{l} \text{ Hicksian demand for} \\ \text{ good } x \end{array} \right]$$

Similarly, by taking the derivatives of eq (iii) with respect to P_y , we get,

$$\frac{\partial e(P, \bar{u})}{\partial P_y} = \frac{\partial}{\partial P_y} \left\{ \left(P_x + \frac{P_y}{2}\right) \bar{u} \right\}$$

$$\therefore h_y(P, \bar{u}) = \frac{1}{2} \bar{u} \quad \left[\begin{array}{l} \text{ Hicksian demand for} \\ \text{ good } y \end{array} \right]$$